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Agentic AI for Early Detection of Lung Cancer in Non-Smokers

Abstract. Early lung cancer detection in never-smokers is an urgent unmet clinical need since its presentation is usually non-specific and there is no guideline for screening. We propose Agentic AI Framework for Explainable Early Lung Cancer Detection in Never-Smokers (ENDN) EFP that combines clinical notes and genomic data for explainable risk pro- filing. This system is based on: a Symptom Extractor Agent which retrieves pertinent symptoms and temporal information from clinical notes, a Symptom-Gene Mapper Agent which relates extracted symptoms to gene expression markers, and a Patient-Gene Scoring Agent which integrates multi-modal data to compute personalized risk scores. The system generates actionable reports for clinicians to facilitate early diagnosis. By leveraging knowledge graph reasoning, the system enables explainability and trust for AI-driven medical decisions. The system was tested on a synthetic data set containing 4,040 never-smoker patient records and the best learned model achieved 85.7% accuracy, 88.3% sensitivity, and 0.867 ROC-AUC, which surpassed baselines by 10-20%. This approach generates interpretable, evidence-based risk profiles to facilitate better outcomes for never-smoker lung cancer patients.

Keywords: Lung Cancer Detection · Agentic AI · Never-Smokers · Knowledge Graphs · Explainable AI · Multi-Agent Systems

1 Introduction

Lung cancer is the most fatal malignancy worldwide, leading to more deaths than other types of cancer. Although tobacco smoking remains the major risk factor, recent epidemiological data show a disturbing pattern: 10-20% of all lung cancers arise in individuals who have never smoked, representing 20,000-40,000 new cases per year in the United States alone [1]. This group is also more likely to be younger and female. Current protocols, based largely on smoking history, actively disfavor never-smokers for enrolment in routine surveillance. The United States Preventive Services Task Force is now advising low-dose computed tomography screening only for persons with significant smoking histories (20-30 pack-years). As a result, never-smokers are diagnosed at later stages when treatment options are limited. This survival gap is striking: localized lung cancer has 60% survival, while metastatic disease has 10-15% survival. Never-smoker lung cancers have distinctive molecular profiles such as higher frequencies of action- able driver mutations including EGFR, ALK and ROS1 fusions. Such molecular alterations are not only predictive of targeted therapy response, but also indicative of distinct disease biology. Deep learning, in particular, enabled medical imaging to relay many subtle attributes that are not visible to human eyes. Neural networks trained on massive data repositories can detect cancer with sensitivity and specificity comparable to expert radiologists. Importantly, these AI systems do not use smoking history as an input variable, allowing for a truly universal risk assessment.

We study the opportunity for early diagnosis of lung cancer in never-smokers in this work by proposing a novel agentic AI framework that meets an urgent demand in the area. Through integration of clinical narratives with NLG and KG-based biological reasoning and ML-based risk prediction, the developed system identifies never-smokers at high risk recommended for screening for LC even though they do not meet the current guideline criteria.

1.1 Problem Definition

The main issue is the underdiagnosis and late diagnosis of lung cancer in never-smokers because of the screening criteria based on the tobacco history alone. This complex problem emerges at various layers:

Limitations of Current Screening Paradigms: Contemporary screening guidelines deny a whole population the benefits of early detection that can save their lives. Never smokers make up 30-40% of lung cancer cases in some populations, such as Asian women and younger individuals. Risk factors for most non-smokers such as residential radon exposure, occupational exposures, air pollution, and genetic predispositions are underappreciated in clinical decision making.

Diagnostic Challenges: Early lung cancer in never-smokers manifests as mild non-specific symptoms that can easily be misinterpreted as symptom of benign disease. Chronic cough, mild dyspnea, pleuritic chest pain, and malaise are common manifestations in both health and non-malignant lung disease. In the absence of heightened suspicion for risk, both patients and clinicians may attribute symptoms to benign causes, resulting in diagnostic delays of several months or years.

Underutilization of Longitudinal Clinical Data: Electronic health records include rich narratives of symptom development and physical examination findings and diagnostic results collected at multiple time points over the years. Early signals of cancer small changes in symptoms or constellations of findings are buried in vast amounts of documentation. With the diverse vocabulary, use of abbreviations, and narrative structures found in clinical notes, pattern recognition is difficult even given sufficient time.

Barriers to Genomic-Clinical Data Integration: Lung cancers in never-smokers are molecularly distinct and exhibit higher rates of actionable driver mutations. While next-generation sequencing (NGS) adoption is becoming widespread, the clinical integration of genomic data is still disjointed. Genomic data often lives in siloed lab systems, removed from clinical documentation and requires manual link up. At present, genomic testing is performed after diagnosis to guide treatment, rather than to enable application of genetic risk information for screening and early detection.

2 Literature Survey

2.1 Epidemiology of Never-Smoker Lung Cancer

The burden of lung cancer in never-smokers is increasing worldwide15 and rates up to 40% of cases in never-smokers have been suggested in some Asian populations predominantly among East Asian females. Adenocarcinoma is the most frequent histological subtype of never-smoker lung cancer, and differs from other histological subtypes in smokers. They are also well-known to harbor therapeutically targetable mutations such as EGFR, ALK and ROS1 fusions which have been associated with better outcomes even at diagnosis, once again underscoring the relevance of early detection. Studies on risk factors have found several environmental and genetic risk factors. Radon, particularly in poorly ventilated homes, is a significant carcinogen associated with a high number of cases annually. Indoor air pollution, secondhand smoke, work exposures (asbestos and diesel exhaust), and ambient air pollution related to traffic and industrial sources are all risks.

2.2 Deep Learning for Medical Imaging Analysis

Convolutional neural networks applied to chest X-rays and CT scans have achieved high success rates in pulmonary nodule identification and lung cancer risk estimation. The Sybil algorithm, developed by Massachusetts General Hospital and MIT, demonstrated lung cancer risk prediction up to six years out from a single low-dose CT scan. Trained on tens of thousands of screening exams from multiple international organizations, Sybil discovered imaging patterns associated with future cancer even without baseline nodules. Importantly, the model performed well in never-smoker risk stratification.

2.3 Natural Language Processing of Clinical Notes

The Electronic health record (EHR) data contain extensive clinical information; however, much information is in the format of free-text narratives. Transformer-based language models (e.g. BERT and its variants) pre-trained on medical text perform very well in clinical entity recognition, relation extraction and temporal information extraction. Recent research shows that language models can learn symptom progression patterns, which predict disease development trajectories.

2.4 Knowledge Graphs in Biomedical Reasoning

Knowledge graphs represent entities and relations so that reasoning can be performed over complex domains. In biomedicine such graphs integrates information from the scientific literature, clinical databases, molecular pathway repositories and ontologies to produce biologically and clinically motivated models of knowledge. Among others, applications such as drug repurposing, adverse event prediction and disease gene prioritization are considered. State-of-the-art work is the integration of knowledge graphs with machine learning models, leveraging biological reasoning and data-driven prediction.

2.5 Multi-Modal Data Integration

Medicine is a practical science as well as an arts and decisions are derived from a variety of information sources: imaging, laboratory testing, clinical documentation, and patient reported outcomes. Studies in [5] demonstrate that the combination of image with clinical and molecular data achieves better prognostic performance than either data source alone in oncology. In lung cancer, combining clinical risk factors, circulating biomarkers, and genomic profiles with CT findings yields more precise risk stratification than imaging-only or clinical-only classification schemes.

2.6 Agentic AI Systems in Healthcare

Agentic systems break down workflows into specialized agents that are responsible for specific tasks and communicate with each other over well-defined interfaces. In medicine, agentic systems are used for clinical decision support, medical question answering, and treatment planning. Chain-of-agents systems that operate sequentially on patient data through a series of agents have been demonstrated to be superior to end-to-end models, in addition, the modular de- sign allows reasoning to be explained via the intermediate representations that are also exposed.

3 Proposed Work

3.1 System Architecture Overview

The designed Multi-Agent AI System for Early Lung Cancer Detection in Never-Smokers consists of a three-phase architecture: Data Collection and Preprocessing, Multi-Agent Process Execution, and Model Validation and Reporting.

Phase 1: Data Collection and Preprocessing Clinical Data Acquisition: Informal language, abbreviations and free text in raw clinical notes are the symptom extraction substrates. The data is organized to capture symptomatic trajectories of never smoking lung cancer cases, with non-specific symptoms such as chronic cough, fatigue, and weight loss.

Genomic Data Integration: The data on gene expression obtained from public repositories (NCBI Gene Expression Omnibus) is limited to lung samples with an equal number of malignant and non-malignant samples. Genomic data are also batch effect corrected and adjusted for study heterogeneity before agent processing.

Phase 2: Multi-Agent Workflow The core of the framework contains three specialized AI agents running one after another:

Agent 1: Symptom and Timeline Extractor - This LLM-based NLP agent extracts symptomatic data from clinical notes. It identifies the presence of symptoms and related attributes (duration, onset, progression), and makes distinctions such as ”dry cough for three months” is not equal to ”new persistent cough.” The output is a structured JSON format including symptom list with temporal information.

Agent 2: Symptom-Gene Mapper - This is powered by our own knowledge graph The KG is a semantic network in which nodes represent symptoms, genes, and biological pathways and edges represent relations. Based on Agent 1’s structured symptom list, it queries the KG for genetic links. For chronic cough and chest pain given as input, it looks for genes and pathways that give rise to these symptoms (EGFR, ALK mutations), and outputs a ranked list of candidate genes.

Agent 3: Patient-Gene Scoring Agent -The generation agent works by collecting outputs of the previous agents to generate the final framework outputs. Takes candidate genes (from Agent 2) and raw patient data and outputs transparent risk scores. The algorithm compares patient symptom history with known clinical trajectories for each candidate gene within the Knowledge Graph using a multi-gene evidence scoring system based on symptom matching, temporality and cross-gene evidence. The output is a list of potential genetic routes ranked with numerical risk scores.

Phase 3: Model Validation and Reporting Performance Validation: Scoring agent output trains and tests baseline machine learning models (Logistic Regression, Random Forest) on structured genomic features to evaluate pipeline improvement, particularly Recall for reducing false negatives.

Clinician-Facing Report Generation: Automated case reports translate numerical risk scores and ranked candidate gene lists into human-readable form, explaining score rationale and symptom contributions to predictions.

3.2 Algorithm Design

Agent 1: Symptom and Timeline Extractor The agent employs LLM capabilities with fine-tuned prompt engineering. Predefined prompts instruct the LLM to identify symptoms and gather duration and quality details. Zero-shot learning enables task performance without large pre-labeled sample tuning. The LLM captures named entities (chest pain, fatigue) and temporal attributes (one year, worsening progression), producing standardized JSON format for inter-agent communication.

Agent 2: Symptom-Gene Mapper Knowledge graph construction forms the agent's main algorithm, with nodes representing Symptom, Gene, and Biological Pathway entities, and edges representing relations . The KG is built by knowledge extraction from the scientific literature, biomedical ontologies (UMLS) and public databases. Graph traversal starts at the patient-specified symptom nodes and path with the strongest evidence-based associations is prioritized. Multi-symptom reinforcement (cough and chest pain both are aligned to EGFR) also produces a higher confidence score.

Agent 3: Patient-Gene Scoring Agent: Agent produces Patient Journey Graph mirroring single patient pathway with symptom timelines of Agent 1a rule-based scoring function computes numeric risk scores for the candidate genes, quantifying the strength of the evidence through:

- Symptom Match: Known gene alteration manifestations
- Time Sensitivity: Symptom duration and progression relevance
- Multi-Symptom Evidence: Shared genetic pathway boosting

The transparent scoring system supports explainability by connecting risk scores with patient symptoms and related genes.

3.3 Dataset and Platform

Dataset Characteristics A synthetic lung cancer dataset with 4,040 patient records was generated using Ollama Llama3 in Docker Desktop allowing for privacy-preserving realistic clinical scenario creation.

Dataset Features:

- Records: 4,040 patients
- Features: 19 columns (demographics, clinical symptoms, outcomes)
- Target: lung cancer (YES: 2,296 cases [56.8
- Age: 25-85 years (mean: 62.67, SD: 8.94)
- Gender: Balanced (Male: 50.27%, Female: 49.73%)
- Smoking: All entries zero (non-smokers)

Clinical Symptoms: All the symptom-related items except Snooze were numerically coded (1=absent, 2=present): Yellow Fingers (mean 1.57), Anxiety (1.50), Peer Pressure (1.50), Chronic Disease (1.51), Fatigue (1.67), Allergy (1.55), Wheezing (1.55), Alcohol Consuming (1.55), Coughing (1.58), Shortness of Breath (1.64), Swallowing Difficulty (1.47), Chest Pain (1.54).

Platform Specifications Development Environment:

- Dataset Generation: Ollama Llama3 in Docker Desktop
- Programming: Python 3.8+
- IDE: Jupyter Notebook, PyCharm, VS Code
- ML Libraries: scikit-learn, pandas, numpy, XGBoost, TensorFlow/PyTorch
- Visualization: matplotlib, seaborn, plotly
- Interpretation: SHAP, LIME

Hardware:

- Processor: Intel Core i5/i7 or AMD Ryzen 5/7
- RAM: Minimum 8GB, recommended 16GB
- Storage: SSD with 20GB available
- GPU: Optional NVIDIA CUDA support
- OS: Windows 10, Linux, macOS

4 Results and Discussion

4.1 Evaluation Metrics

The detection results were evaluated via standard classification metrics, that is, overall correctness in terms of accuracy, true positive rate (TPR) equivalent to sensitivity, true negative rate (TNR) equivalent to specificity, positive predictive value (PPV) comparable to precision, and negative predictive value (NPV). ROC-AUC represented the overall ability to discriminate between subjects above and below the disease threshold. F1 score was a harmonic mean of precision and recall, important in the context of low disease prevalence.

4.2 Framework Performance

Agent 1 Performance

- Precision: 84.2% for symptom entity recognition
- Recall: 81.7% for documented symptom capture
- F1-Score: 82.9% balanced extraction performance
- Temporal Attribute Extraction: 78.5% accuracy

Agent 2 Performance

- Mapping accuracy: 87.3% true gene associations recovered
- Pathway Coverage: 92.1% known lung cancer pathways
- Average Candidate Genes: 3.4 genes per patient
- Biologically Plausible Mappings: 94.6% expert validation

Agent 3 Performance

– Risk Score Accuracy: 88.1% concordance with diagnoses
– High-Risk Identification: 91.4% true positives flagged
– Explanation Quality: 4.2/5.0 clinical reviewer rating
– Processing Time: Average 4.3 seconds per record

4.3 End-to-End System Performance

Table 1. System Performance Metrics

Metric	Value
Overall Accuracy	85.7%
Sensitivity (Recall)	88.3%
Specificity	82.1%
Precision (PPV)	86.9%
Negative Predictive Value	84.2%
F1-Score	87.6%
ROC-AUC	0.867

The high sensitivity (88.3%) is particularly encouraging for cancer detection since it identifies most true lung cancers. The balanced F1 score of 87.6% suggests that the false positive rate and false negative rate are well balanced.

4.4 Baseline Comparison

Table 2. Comparison of performance with baseline methods

Method	Accuracy	Sensitivity	ROC-AUC
Agentic AI (Proposed)	**85.7%**	**88.3%**	**0.867**
Random Forest	81.2%	79.6%	0.823
XGBoost	82.9%	82.1%	0.841
Logistic Regression	78.4%	75.3%	0.791
Clinical Risk Scores	72.6%	68.9%	0.742

Our agentic AI paradigm outperforms these methods on all metrics, achieving a 10-20% gain over clinical risk scoring baselines.

Table 3. Symptom Prevalence and Feature Importance

Symptom	Prevalence	Importance
Fatigue	67%	0.189
Shortness of Breath	64%	0.176
Coughing	58%	0.162
Yellow Fingers	57%	0.143
Chest Pain	54%	0.128
Wheezing	55%	0.114
Alcohol Consuming	55%	0.088

4.5 Symptom Analysis

Fatigue with breathlessness and cough Accuracy of these symptoms in predicting lung cancer is similar to that previously reported. SHAP feature importance analysis validated these symptoms as the largest risk prediction contributors, further supporting their clinical feasibility.

4.6 Temporal Pattern Recognition

The system performed very well in tracking temporal evolution of symptoms:

- Progressive Symptom Detection: 82.7% accuracy
- Symptom Clustering: 73.4% multiple symptoms identified in 6-month windows
- Long-Term Risk Estimation: 12.3% better accuracy for predictions over the period of 12 or more months as compared to single-visit risk estimation

4.7 Clinical Utility Assessment

Five oncologists appraised system outputs via semi-structured interviews:

Table 4. Clinician Assessment Scores (Scale 1-5)

Assessment Criteria	Mean Score
Explanation Clarity	4.2
Clinical Relevance	4.4
Trustworthiness	4.0
Decision Support Usefulness	4.3
Practice Change Potential	4.1
Overall Clinical Utility	**4.2**

Clinician rated all domains positively, particularly in terms of clinical relevance and IUS, suggesting the potential to be beneficial when applied in reality.

4.8 Key Findings

Strengths:

- Explainability by knowledge graph-based reasoning
- Temporal modeling capturing the evolution of symptoms
- Integration of clinical text and genomic data at the multimodal level
- Clinical coordination with medical thinking

Limitations:

- Synthetic data must still be validated in the real clinical setting
- The computational resources that are needed
- The requirements for maintaining the knowledge graph
- Generalization to different populations

5 Conclusion and Future Work

In this study, we formulated and validated a novel multi-agent AI system for predicting lung cancer in never-smokers, which is a major void in the clinical practice. The three-agent pipeline enables end-to-end analysis of unstructured clinical notes up to full risk profiling, with a performance of 85.7% accuracy, 88.3% sensitivity, and 0.867 ROC-AUC.

The main technical novelty of our method is that it considers both unstructured clinical notes and structured genomic data in a unified manner in the form of explainable knowledge graph reasoning. This method achieved significantly better results than the baseline approaches (an improvement of 10-20%) and was considered clinically interpretable by oncologists (4.2/5.0).

5.1 Contributions

Methodological: We present a multi-agent architecture that combines temporal symptom modeling and knowledge graph representation in a joint explainable AI framework.
Technical: NLP for clinical notes, graph-based reasoning algorithms, Rule- based scoring with clinical interpretation, and automated end-to-end pipeline.
Clinical: Symptoms affected at least 50% of participants (fatigue 67%, dyspnea 64%, cough 58%), proved to effectively stratify risk, and were externally validated by clinicians with high scores.

5.2 Future Enhancements

Clinical Validation: Multi-center cooperation for real clinical data validation, prospective studies, heterogeneous population inclusion and long-term outcome follow-up.

Technical Improvements: Domain-specific fine-tuning, automated knowledge graph updating, medical imaging support, uncertainty quantification, and scalability enhancement.

Extended Applications: Customization for other malignancies, long-term disease surveillance, therapeutic response forecasting, and identification of side effects.

The success of this demonstration shows the promise of agentic AI for early cancer detection in an underserved group of never-smokers, and sets the stage for tackling general medical AI problems.